

CAPE BYRON LH, Australia

WMO: 945990

Lat: 28.633S

Lon: 153.633E

Elev: 98

StdP: 100.15

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	11.2	-1.7	3.3	14.0	0.2	3.9	14.3	14.8	17.1	13.2	16.7	7.7	240	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	29.5	23.3	28.4	22.9	27.5	22.6	24.9	27.3	24.2	26.7	23.7	26.2	5.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	19.4	26.0	23.6	18.6	25.6	22.9	17.9	25.2	76.6	27.2	73.8	26.7	71.5	26.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.7	11.2	DB	8.6	31.9	0.8	1.0	8.0	32.6	7.5	33.2	7.0	33.7	6.5	34.5	
			WB	4.5	26.1	0.9	1.1	3.8	26.9	3.3	27.5	2.7	28.1	2.1	28.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	24.4	24.1	23.2	20.9	18.2	16.2	15.6	16.5	18.4	19.9	21.6	23.0
	DBStd	3.56	1.66	1.73	1.75	1.81	1.84	1.88	1.97	2.04	2.00	1.93	1.80	1.85
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	282	0	0	0	2	26	68	88	64	23	8	2	1
	CDD10.0	3701	446	396	411	326	253	185	173	201	253	308	347	402
	CDD18.3	941	187	162	152	78	21	4	2	7	26	58	99	144
	CDH23.3	3401	988	842	581	129	11	0	1	4	22	77	227	518
Wind	WSAvg	6.2	6.3	6.7	6.3	6.2	5.6	6.1	5.7	5.9	6.2	6.5	6.3	6.4
	PrecAvg	1730	159	176	217	192	197	162	97	86	58	111	129	153
	PrecMax	2476	444	409	652	611	547	412	329	276	281	566	347	426
	PrecMin	1126	30	40	30	45	19	16	1	9	0	8	18	42
	PrecStd	390	100	81	142	115	117	108	76	69	58	92	79	96
	DB	31.2	30.8	29.6	26.9	24.6	22.5	22.2	23.9	25.7	26.9	28.1	30.1	
	MCWB	24.4	23.8	23.8	22.0	19.5	17.8	16.8	15.9	17.3	19.9	21.6	22.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	29.6	29.1	28.2	25.7	23.3	21.2	21.0	21.9	23.5	25.2	26.7	28.5	
	MCWB	23.5	23.4	23.0	21.2	18.7	17.2	15.9	16.5	17.3	19.4	20.7	22.3	
	DB	28.3	28.0	27.2	24.8	22.3	20.2	20.0	20.9	22.5	24.0	25.5	27.1	
	MCWB	23.0	23.3	22.4	20.7	18.1	16.6	15.8	16.1	17.4	19.2	20.5	22.0	
	DB	27.3	27.1	26.1	23.9	21.4	19.3	19.0	19.9	21.6	23.1	24.6	26.0	
	MCWB	22.7	22.9	22.0	20.2	17.4	15.9	15.3	15.6	17.2	18.9	20.3	21.6	
	WB	25.9	26.4	25.1	24.0	21.9	19.9	19.2	19.3	20.8	22.4	23.6	24.9	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	28.5	27.8	27.7	25.3	23.0	20.8	20.5	20.9	22.1	23.8	26.0	27.2	
	WB	24.8	25.0	24.1	22.7	20.0	18.5	18.2	19.7	21.4	23.1	24.7	26.5	
	MCDB	27.4	27.0	26.6	24.5	22.3	20.1	19.7	20.5	21.4	23.1	24.7	26.5	
	WB	24.2	24.2	23.4	21.7	19.0	17.6	17.1	17.4	19.0	20.6	21.7	23.2	
	MCDB	26.6	26.7	25.8	23.7	21.2	19.4	19.0	19.9	21.0	22.6	23.9	25.5	
	WB	23.6	23.5	22.8	20.8	18.1	16.7	16.0	16.5	18.2	19.9	21.2	22.5	
	MCDB	26.0	26.2	25.2	23.2	20.5	18.6	18.2	19.2	20.6	22.1	23.4	24.8	
Mean Daily Temperature Range	MDBR	5.6	5.4	5.2	5.3	5.4	5.2	5.6	6.0	5.8	5.8	5.8	5.6	
	5% DB	7.3	6.8	6.6	6.1	6.1	5.7	6.5	6.8	6.9	7.4	7.3	7.5	
	MCWBR	3.5	3.4	3.4	3.6	4.2	4.2	4.7	4.9	4.6	4.5	3.7	4.1	
	5% WB	6.2	6.0	5.9	5.6	5.4	5.0	5.4	6.0	5.5	6.3	6.0	6.2	
	MCWBR	3.5	3.2	3.6	3.8	3.9	4.2	4.5	4.9	3.9	4.1	3.5	3.7	
	taub	0.384	0.370	0.352	0.338	0.316	0.310	0.301	0.314	0.349	0.378	0.375	0.385	
	taud	2.523	2.575	2.622	2.636	2.654	2.662	2.675	2.620	2.514	2.458	2.510	2.505	
Clear-Sky Solar Irradiance	Ebn at Noon	958	955	941	908	885	868	893	920	928	935	960	959	
	Edh at Noon	113	104	94	84	75	70	72	84	103	115	113	115	
	RadAvg	6.50	5.91	5.02	4.21	3.47	2.87	3.34	4.23	5.26	5.91	6.44	6.45	
	RadStd	0.51	0.46	0.27	0.21	0.21	0.18	0.27	0.33	0.42	0.51	0.50	0.65	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-58	+171	+105

Nomenclature: See separate page